WORKED SOLUTION

STEP I 1987, Question 6

Pure • Areas and Inverse Functions

STEP I 1987, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q6

The key idea

$\mathrm{F}(s)$ is the area *under* the curve, between it and the x-axis; $\mathrm{G}(t)$ is the area to the *left* of the curve, between it and the y-axis. Drawn on the same picture, these two regions together cover the rectangle with corners $(0,0)$ and (s,t), with something to spare unless the corner (s,t) sits exactly on the curve. For the second part we choose s to make that happen, and then the hard integral $\int \sin^{-1} y \,\mathrm{d}y$ comes for free from the easy one, $\int \sin x \,\mathrm{d}x$.

Solution

The inequality $(*)$

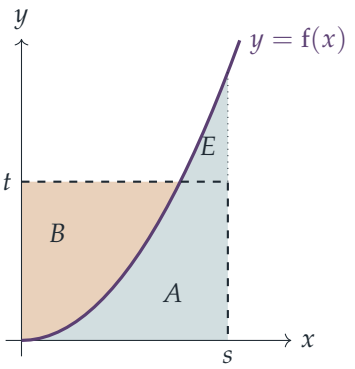
First, a fact about f. It is continuous and takes each value at most once, so it is strictly monotonic (a continuous function that went up and then down would take some value twice, by the intermediate value theorem). Since $\mathrm{f}(0) = 0$ and $\mathrm{f}(x) \geq 0$, it must be strictly *increasing*. So g is increasing too, and for $0 \leq x \leq a$, $0 \leq y \leq \mathrm{f}(a)$,

$$y \leq \mathrm{f}(x) \iff \mathrm{g}(y) \leq x.$$

Define two regions:

$$A = \{(x,y) : 0 \leq x \leq s,\ 0 \leq y \leq \mathrm{f}(x)\}, \qquad \text{area } \mathrm{F}(s),$$
$$B = \{(x,y) : 0 \leq y \leq t,\ 0 \leq x \leq \mathrm{g}(y)\}, \qquad \text{area } \mathrm{G}(t),$$

and let R be the rectangle $0 \leq x \leq s$, $0 \leq y \leq t$, of area st. The area of B is $\mathrm{G}(t)$ because we are integrating horizontal strips of length $\mathrm{g}(y)$ ("area with respect to y").



Every point of R lies in A or in B: if $(x, y) \in R$ then either $y \le \mathrm{f}(x)$, so the point is in A, or $y > \mathrm{f}(x)$, which means $x < \mathrm{g}(y)$, so the point is in B. The overlap $A \cap B$ is just the arc of the curve itself, which has zero area. Hence

$$\mathrm{F}(s) + \mathrm{G}(t) = \text{area}(A) + \text{area}(B) \ge \text{area}(A \cup B) \ge \text{area}(R) = st,$$

which is $(*)$.

Ask yourself Exactly what has been counted that isn't in the rectangle?

The excess is the part E of $A \cup B$ sticking out of R. If $t < \mathrm{f}(s)$ (as in the figure), B stays inside R, but A pokes out above $y = t$ in the region $\mathrm{g}(t) < x \le s$, $t < y \le \mathrm{f}(x)$. Since $\mathrm{f}(x) > t$ for every $x > \mathrm{g}(t)$ and f is continuous, this region has positive area, so the inequality is strict. If $t > \mathrm{f}(s)$, the same happens with the roles swapped: B pokes out to the right of $x = s$. If $t = \mathrm{f}(s)$, the corner (s, t) lies on the curve, $A \cup B$ is exactly R, and there is no excess.

So **equality holds in $(*)$ if and only if** $t = \mathrm{f}(s)$, that is, when (s, t) lies on the curve.

The integral of $\sin^{-1} y$

Take $\mathrm{f}(x) = \sin x$ on $[0, \frac{1}{2}\pi]$. It is continuous, increasing, passes through the origin and stays in the first quadrant, so everything above applies with $a = \frac{1}{2}\pi$, $\mathrm{f}(a) = 1$ and $\mathrm{g}(y) = \sin^{-1} y$.

Given $t \in [0, 1]$, choose $s = \sin^{-1} t$, which lies in $[0, \frac{1}{2}\pi]$. Then $t = \sin s = \mathrm{f}(s)$, so equality holds in $(*)$:

$$\int_0^t \sin^{-1} y \, \mathrm{d}y = \mathrm{G}(t) = st - \mathrm{F}(s).$$

Now the easy integral:

$$\mathrm{F}(s) = \int_0^s \sin x \, \mathrm{d}x = 1 - \cos s.$$

Substituting $s = \sin^{-1} t$,

$$\int_0^t \sin^{-1} y \, \mathrm{d}y = t \sin^{-1} t - (1 - \cos(\sin^{-1} t)),$$

as required.

Checking by differentiation

Since $\sin^{-1} t \in [0, \frac{1}{2}\pi]$, its cosine is non-negative, so $\cos(\sin^{-1} t) = \sqrt{1-t^2}$. The right-hand side is therefore $t\sin^{-1} t - 1 + \sqrt{1-t^2}$. For $0 \le t < 1$, using $\dfrac{\mathrm{d}}{\mathrm{d}t}\sin^{-1} t = \dfrac{1}{\sqrt{1-t^2}}$ and the product rule,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left(t\sin^{-1} t - 1 + \sqrt{1-t^2}\right) &= \sin^{-1} t + \frac{t}{\sqrt{1-t^2}} - \frac{t}{\sqrt{1-t^2}} \\ &= \sin^{-1} t,\end{aligned}$$

which, by the fundamental theorem of calculus, is the derivative of the left-hand side. Both sides are 0 at $t = 0$, so they agree on $[0, 1)$, and by continuity at $t = 1$ as well.

Checks

- At $t = 1$: the formula gives $\frac{1}{2}\pi - 1 \approx 0.5708$. Geometrically, the area left of $y = \sin x$ up to height 1 is the rectangle $\frac{1}{2}\pi \times 1$ minus the area under the arch, 1. A numerical integration of $\sin^{-1} y$ over $[0, 1]$ gives 0.57080.
- At $t = \frac{1}{2}$: the formula gives $\frac{\pi}{12} - 1 + \frac{\sqrt{3}}{2} \approx 0.12782$, matching Simpson's rule.
- For $(*)$, try $\mathrm{f}(x) = x^2$, $s = 1$, $t = 4$: $\mathrm{F} + \mathrm{G} = \frac{1}{3} + \frac{16}{3} = \frac{17}{3} > 4$, while with $t = 1 = \mathrm{f}(1)$ we get $\frac{1}{3} + \frac{2}{3} = 1 = st$.

Common slips

- Writing $\cos(\sin^{-1} t) = \pm\sqrt{1-t^2}$ without choosing the sign: the range of $\sin^{-1}$ makes it positive.
- Saying only "the regions cover the rectangle" without explaining why the inequality is strict when $t \ne \mathrm{f}(s)$; the equality question needs that.
- Forgetting that f must be increasing: the picture relies on it.

Beyond STEP

The inequality $(*)$ is *Young's inequality*. With $\mathrm{f}(x) = x^{p-1}$ it becomes $st \le \frac{s^p}{p} + \frac{t^q}{q}$ where $\frac{1}{p} + \frac{1}{q} = 1$, the key step in proving Hölder's inequality, one of the workhorses of analysis. The pairing of F and G is an early glimpse of the Legendre transform, which links Lagrangian and Hamiltonian mechanics. For the analysis behind all this, try Bartle and Sherbert, *Introduction to Real Analysis*.

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